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Introduction

1. INTRODUCTION

Physical systems are designed and built to perform certain defined functions. Submarines, aircraft, and spacecraft must navigate in their respective environments to accomplish their objectives, whereas an electric power system network must meet the load demands. In order to determine whether a system is performing properly, and ultimately to control the system performance, the engineer must know what the system is "doing" at any instant of time. In other words, the engineer must know the *state* of his system. In navigation, the state consists of position and velocity of the craft in question; in an ac electric power system, the state may be taken as voltages and phase angles at network nodes. Physical systems are often subject to random disturbances, so that the system state may itself be random. In order to determine the state of his system, the engineer builds a measuring device (e.g., sextant, radar, voltmeter) and takes *measurements* or *observations* on his system. These measurements are generally contaminated with noise caused by the electronic and mechanical components of the measuring device.

The problem of determining the state of a system from noisy measurements is called *estimation*, or *filtering*, and is the main subject of this book. It is of central importance in engineering, since state estimates are required in the monitoring, and for the control of systems. Furthermore,

a large class of (system) identification problems can be regarded as problems of filtering.

This book takes the state-space approach to filtering. The physical (dynamical) system is modeled by a finite-dimensional Markov process, the output of a stochastic (random) differential or difference equation. The quasi-steady state estimation in an ac electric power system network is an example of a static problem, which is a special case of the dynamical model system representation used in this book. We take the Bayesian, or probabilistic, point of view in filtering. That is, we look upon the conditional probability density function of the state, given the measurements, as the fundamental entity in the problem. This conditional density embodies all the information about the state of the system which is contained in the available measurements, and all estimates of the state can be constructed from this density.

Our approach enables us to give a rather rigorous and relatively complete and self-contained treatment of nonlinear, as well as linear, filtering theory. Yet we avoid the measure-theoretic aspects of probability theory, thus making our work accessible to engineers. In fact, this book is principally intended for engineers. Because mathematical rigor has not been completely sacrificed, our work should also be of interest to the applied mathematician.

2. SCOPE AND OBJECTIVES

Following the fundamental work of Kalman and Bucy [3, 4][1] in linear filtering theory, a host of papers and reports appeared, formally deriving their linear filtering algorithm via "least squares," "maximum likelihood," and other classical, statistical methods.[2] These statistical methods were also formally applied to the nonlinear estimation problem. Using linearization of one sort or another, Kalman-like filtering algorithms were developed and applied to nonlinear problems. This statistical work received its impetus from the aerospace dollar. Work was duplicated, triplicated; everyone derived his own Kalman filter, perhaps partly because of a lack of understanding of Kalman's original work.

[1] Numbers in brackets refer to references or bibliography at the end of each chapter.

[2] A sketch of the developments in filtering theory, including the earlier work of Wiener, may be found in Chapter 5, especially in Sections 5.1 and 5.3.[3]

[3] Section 5.1 refers to Section 1 of Chapter 5. We use the following convention. In referring to a section of another chapter, the section number is preceded by the chapter number, as in Section 5.1. When referring to a section of the same chapter, only the section number is used. Thus if we refer to Section 1 here, we mean Section 1 of the present chapter.

In the meantime, such important aspects of the linear theory as filter stability, error sensitivity, and the problem of modeling, some of which were treated in Kalman's original work, went largely ignored. It is a truism, however, that the step from theory to practice is a large and difficult one, and that the aerospace community contributed much to the reduction of the theory to practice. The statistical work served to bring linear filtering within the grasp of the practicing engineer, and led to the applications[4] of the theory in satellite orbit determination, submarine and aircraft navigation, and space flight, including the Ranger, Mariner, and Apollo missions. On the other hand, the statistical methods tend to mask the fundamental probabilistic structure of the filtering problem. This structure consists of a stochastic process evolving in time, whose probability law is to be determined from observations of another, related, stochastic process. It becomes increasingly important to exploit this probabilistic structure when dealing with the nonlinear filtering problem.

While Kalman and Bucy were formulating the linear theory in the United States, Stratonovich [9] was pioneering in the development of the probabilistic approach to nonlinear filtering in Russia. The work of Stratonovich was not immediately known in the West, however, and Kushner [5, 6] and also Wonham [10] independently developed the continuous nonlinear theory here. Ho and Lee [1] and Jazwinski [2], respectively, applied the probabilistic approach to discrete and continuous-discrete[5] nonlinear problems. Much of the subsequent theoretical work in nonlinear filtering was done by Kushner.

Most of the work in nonlinear filtering is very theoretical, involving such hitherto obscure and difficult subjects as "stochastic differential equations" and the "Itô calculus," which require a fair facility with measure theory for understanding. An engineer interested in applications is virtually lost. Furthermore, it is not clear from the literature whether the nonlinear theory can be fruitfully applied to any significant practical problem. A student or engineer interested in learning the linear theory also has a difficult task before him. There is a large body of journal literature on the subject which thus far has not been collected in a single source.

Our objective in writing this book is twofold. First, it is to present the current state-of-the-art in nonlinear filtering theory in a form accessible to engineers, yet without completely sacrificing mathematical rigor. This includes the development of approximate nonlinear filters suitable

[4] See Chapter 8 for aerospace applications.

[5] Problems with continuous dynamics, but with discrete observations.

for applications, and an analysis of their efficacy in nonlinear problems. Second, it is to gather together, in a unified presentation, many of the results in linear filtering theory. This includes such important properties of linear filters as stability and model error sensitivity. We emphasize the applications of the linear theory to nonlinear problems, and treat the current problem area of filter divergence.

We have attempted to make our book relatively complete and self-contained. To this end, we have included essentially all the material on probability theory and stochastic processes we need for our presentation of filtering theory. This includes a thorough treatment of stochastic differential equations. We avoid measure theory by using mean square convergence rather than probability one convergence. The background we require of the reader consists essentially of advanced calculus, theory of ordinary differential equations, and matrix analysis, plus some mathematical sophistication. Aside from mean square calculus and stochastic differential equations, which we develop fully in this book, we use probability theory and stochastic processes on an engineering level comparable to that in Parzen [8] or Papoulis [7].

The unifying theme in the book is the probabilistic or Bayesian approach. We develop the nonlinear theory first, and obtain linear filtering algorithms as special cases of the nonlinear results. We do, however, include many of the more popular derivations of linear filters in examples. Thus a reader interested only in the linear theory and its applications can avoid much of the mathematics, as we explain in Section 3. Our book is thus suitable as a text for a full two-semester graduate course in filtering theory (linear and nonlinear), or for a one-semester graduate course in linear filtering theory. We include a number of examples that illustrate the concepts. These can be supplemented by the instructor or the reader himself.

3. A GUIDED TOUR

We describe here the contents of our book and indicate how readers, with varying interests and mathematical backgrounds, might approach it.

Background material in probability theory and stochastic processes is given in Chapters 2 and 3. Most of this material is on the engineering level comparable to Parzen [8] or Papoulis [7], with the following exceptions. Certain properties of moments and important inequalities are developed in Section 2.4. That same section also contains an introduction to (random) vectors and matrices. The concepts of convergence of random sequences are defined in Section 3.3, and mean square calculus is developed in Section 3.4. These are required for the study of stochastic

differential equations and stochastic calculus in Chapter 4, which in turn are required for the development of the continuous nonlinear filter in Chapter 6. Section 3.9 is devoted to a study of stochastic difference equations, which is prerequisite for discrete filtering.

The reader familiar with probability theory and stochastic processes on the engineering level might, with the exceptions just described, use Chapters 2 and 3 for reference or review. We would advise the review of Section 2.5 on conditional probabilities and expectations, Section 2.6 on Gaussian random variables, and Section 3.8 on white noise. We might add that our treatment of probability theory and stochastic processes is restricted to continuous random variables and stochastic processes with a continuous state space. Furthermore, we deal only with Markov processes. These are the processes generated by stochastic difference and differential equations; the models we use for dynamical systems.

Chapter 4 is devoted to a study of stochastic differential equations. The theme here is the modeling of a certain stochastic process as the solution of a (stochastic) differential equation. This leads to definition of the Itô stochastic integral and proof of existence and uniqueness of solutions of stochastic differential equations. It is here that we use the mean square calculus developed in Section 3.4. This is followed by a formal derivation of Itô's lemma, which is a fundamental tool in continuous nonlinear filter theory. We next introduce another stochastic integral, that of Stratonovich, and show its relationship with the Itô integral. We then deal, insofar as our mathematical tools permit us, with the important problem of modeling a physical process by a stochastic differential equation. We use the Itô integral throughout the book, and our justification for so doing is given in Section 4.10. Kolmogorov's equation is derived in Section 4.9. It describes the evolution of the unconditioned probability density function of our process and thus is a fundamental tool in continuous nonlinear filtering theory.

Chapter 5 begins the study of the main subject of our book by introducing and defining the problems of filtering, prediction, and smoothing. Our probabilistic approach is described, with emphasis on optimality and optimality criteria in estimation. We also describe the statistical approach to filtering and its probabilistic interpretations.

Chapter 6 presents the (mathematical) solutions to nonlinear filtering problems. We develop equations of evolution for the conditional probability density function, for moments, and for the mode, in continuous, continuous-discrete, and discrete filtering and prediction problems. We also develop the concept of limited memory filtering, with a view toward applications in problems with erroneous system models.

The nonlinear theory is specialized to linear problems in Chapter 7, and linear filters and predictors are developed. More popular derivations of the linear filters, using nonprobabilistic methods, are included in examples. We then introduce the concepts of information, observability, and controllability, and use these in a study of error bounds and stability of the linear filters. The emphasis here is on the discrete filter, since this has not been explicitly treated in the literature. We then study the sensitivity of filter performance to errors in the statistical and dynamical system model. Finally, the linear limited memory filter is developed.

The last two chapters deal with applications.

Chapter 8 develops the applications of linear filter theory, particularly to nonlinear problems. Here we emphasize the discrete and continuous-discrete problems, and go into considerable detail to assist the reader in getting a firm grasp on the theory, by applying the filter to a concrete example problem. We then present actual applications of linear filters in orbit determination and reentry. The balance of the chapter is devoted to the problem of filter divergence due to model errors, and techniques found useful in preventing divergence. We present simulations of two such techniques: the adaptive and limited memory filters.

Chapter 9 is devoted to the development, analysis, and simulations of approximate nonlinear filters. We critically assess the state-of-the-art in nonlinear filtering, and suggest some areas for future research.

Readers who are primarily interested in linear filtering, or those who either do not have the mathematical background required for the nonlinear theory and/or prefer to study the linear theory first, might proceed as follows. Find any of a number of derivations of the linear filtering algorithm via nonprobabilistic methods in the examples of Chapter 7 (Examples 7.1–7.4 for the discrete filter; Examples 7.9–7.13 for the continuous filter). Then proceed to a study of the linear theory in the remainder of Chapter 7, and its applications in Chapter 8. This route, of course, still requires an engineering background in probability and stochastic processes.

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